

PATH-SPACE COTANGENT RIGIDITY, UNIVERSAL CONTRACTIONS, AND TANGENT REPRESENTATIONS

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ABSTRACT. We separate weighted strict Radon duality, strong spectral response, and closed form realizations. All varying Hilbert spaces are transported to a fixed reference space and the metric connection is included in every response derivative. Dissipativity makes the finite-dimensional compressed resolvent strictly accretive in the right half-plane; resolved vectors outside the generator domain are handled by a form-level Schur complement. We prove model-specific pressure-functional rigidity, changing-filtration optional projection convergence, and a category whose typed contractions preserve rates, covariances, cotangents, and memory.

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CONTENTS

1. Round-twenty-one positive closure: weighted strict duality, verified form transport, and innovation-stable contractions	2
1.1. The weighted strict topology and its dual	2
1.2. Verified type-(B) form transport	3
1.3. Form compression and memory connection	4
1.4. Model-specific pressure rigidity	5
1.5. Changing filtrations and the innovation representation	6
1.6. A nonvacuous contraction calculus	7
Scope and verification status	8

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1. ROUND-TWENTY-ONE POSITIVE CLOSURE: WEIGHTED STRICT DUALITY, VERIFIED FORM TRANSPORT, AND INNOVATION-STABLE CONTRACTIONS

This section replaces the former Round-Seventeen source. The weighted strict topology is defined by explicit seminorms. Coboundary separation is stated only for the actual platforms, where an invariant weighted probability has already been constructed. Form transport, rigidity, changing-filtration limits, and contractions are proved separately rather than inserted into the definition of a morphism.

1.1. The weighted strict topology and its dual. Let E be a Polish space and $W : E \rightarrow [1, \infty)$ a continuous proper weight. Put

$$C_W(E) = \{f \in C(E) : \|f/W\|_\infty < \infty\}.$$

For every nonnegative $\psi \in C_0(E)$ define

$$p_\psi(f) = \|\psi f\|_\infty,$$

and for $R > 0$ define

$$q_R(f) = \sup_{W(x) > R} \frac{|f(x)|}{W(x)}.$$

The weighted strict topology β_W is generated on each norm-bounded set by all p_ψ together with the requirement $q_R(f) \rightarrow 0$ uniformly as $R \rightarrow \infty$. Equivalently, a net $f_i \rightarrow 0$ when $\sup_i \|f_i/W\|_\infty < \infty$, $f_i \rightarrow 0$ uniformly on every compact set, and its normalized tails vanish uniformly.

Let

$$\mathcal{M}_W(E) = \left\{ \mu : \int W d|\mu| < \infty \right\}.$$

Theorem 1.1 (Exact weighted strict dual). *Every β_W -continuous linear functional on $C_W(E)$ is uniquely of the form*

$$L(f) = \int f d\mu, \quad \mu \in \mathcal{M}_W(E),$$

and conversely every such measure is continuous. On norm-bounded subsets, β_W is the Mackey topology of the dual pair $(C_W(E), \mathcal{M}_W(E))$.

Proof. Let $K_n = \{W \leq n\}$. Continuity gives a neighborhood

$$|L(f)| \leq 1 \quad \text{whenever} \quad p_{\psi_j}(f) < \varepsilon, \quad q_R(f) < \varepsilon$$

for finitely many ψ_j and one R . On functions supported in K_R , the supremum norm is controlled by these seminorms after enlarging the compact support of the ψ_j . Riesz representation therefore gives a finite measure μ_R on K_R . Uniqueness makes these measures compatible as R increases, so they define one countably additive Borel measure μ .

To obtain the global weighted bound, choose disjoint compact annuli and continuous functions f_n supported there with $|f_n| \leq W$ and sign equal to the local polar sign of μ . If $\int W d|\mu| = \infty$, finite signed sums of the f_n have uniformly bounded W -norm, converge to zero on compacts after shifting to

the tail, and have q_R tending to zero, but $|L|$ is unbounded. This contradicts continuity. Hence $\mu \in \mathcal{M}_W(E)$.

Conversely, split the integral over K_R and its complement. Uniform convergence on K_R controls the first part, while $q_R(f_i) \int_{K_R^c} W d|\mu|$ controls the second. This proves continuity. Uniform integrability characterizes weakly compact subsets of $\mathcal{M}_W(E)$, and the seminorm description is uniform convergence on those subsets, giving the Mackey statement. $\square$

Suppose now that Θ_t is either the A4 history suspension or the B4 kinetic semigroup trajectory map. In both cases there is an invariant probability $\pi \in \mathcal{M}_W(E)$ and $W(\Theta_t x) \leq C_t e^{ct} W(x)$.

Corollary 1.2 (Coboundary annihilator). *The closed weighted-strict span*

$$\mathcal{C} = \overline{\text{span}}\{g - g \circ \Theta_t : g \in C_W(E), t \geq 0\}^{\beta_W}$$

has annihilator equal to the invariant measures in $\mathcal{M}_W(E)$. In particular, a nonzero constant is not in $\mathcal{C}$.

Proof. By Theorem 1.1, annihilation of every coboundary is exactly invariance of the representing measure. The actual platform invariant probability π annihilates $\mathcal{C}$ but integrates a nonzero constant to a nonzero number. No existence assertion is made for an arbitrary Polish flow. $\square$

1.2. Verified type-(B) form transport. Let $H_\eta = L^2(E, \pi_\eta)$ be a family of Hilbert realizations and let $U_\eta : H_0 \rightarrow H_\eta$ be multiplication by the square root of the positive Radon–Nikodym density followed, when needed, by the A4/C1 coordinate transport. Let $V \subset H_0$ be the transported common form domain.

For the A4 suspension, V is the flight H^1 domain with the Doob boundary trace. For the B4 tangent platform, V is the graph form domain of B3. The transported forms are denoted $a_\eta : V \times V \rightarrow \mathbb{C}$.

Theorem 1.3 (Platform type-(B) theorem). *On compact regular parameter/source charts:*

- (i) V is independent of η and dense in H_0 ;
- (ii) for some ω, c, C ,

$$\text{Re } a_\eta(u, u) + \omega \|u\|_{H_0}^2 \geq c \|u\|_V^2, \quad |a_\eta(u, v)| \leq C \|u\|_V \|v\|_V;$$

- (iii) $\eta \mapsto a_\eta$ is C^2 in $\mathcal{L}(V, V^*)$, including the derivative of the Hilbert metric and of U_η ;
- (iv) the associated generators L_η are maximal dissipative after the fixed shift, and

$$\nabla_\eta(z - L_\eta)^{-1} = (z - L_\eta)^{-1}(\nabla_\eta L_\eta)(z - L_\eta)^{-1}$$

as a map $V^* \rightarrow V$.

Proof. For A4, the fixed collision charts identify the flight derivative and the Doob boundary trace. The positive eigenfunction and its inverse are bounded multipliers on the fixed A4 weighted space; hence multiplication transports the boundary condition without changing V . Flight integration by parts, the A4 spectral drift, and the trace estimate give the sectorial coercivity. The source and geometric derivatives are bounded by the A4 fixed-space analytic theorem.

For B4, B3's graph observability gives coercivity on the tangent quotient, and the positive multiplier transfer identifies nearby measures and collision forms on one graph domain. The logarithmic density derivatives are bounded on the regular chart, so the metric connection and form derivatives lie in $\mathcal{L}(V, V^*)$. The representation theorem for closed sectorial forms gives the associated generators. Differentiate the weak equation

$$a_\eta(R_\eta f, v) + z(R_\eta f, v) = (f, v)$$

in V^* ; coercivity justifies inversion and yields the displayed formula with all terms typed in V, V^* . $\square$

1.3. Form compression and memory connection. Let $\mathcal{R} \subset V$ be finite dimensional and let P be the H_0 -orthogonal projection. Assume, as verified for the A4 resolved spaces, that $P : V \rightarrow V$ is bounded. Put $Q = I - P$. For $\operatorname{Re} z > 0$, solve the complementary form equation

$$a_\eta(w, Qv) + z(w, Qv) = -a_\eta(r, Qv), \quad v \in V, \quad w \in QV.$$

Coercivity defines $w = W_\eta(z)r$. The Schur form on $\mathcal{R}$ is

$$s_\eta(z)[r, r'] = (zr, r') + a_\eta(r + W_\eta(z)r, r').$$

Theorem 1.4 (Form Schur complement and covariant derivative). *The matrix $S_\eta(z)$ representing $s_\eta(z)$ is invertible for $\operatorname{Re} z > 0$ and equals the inverse of the compressed resolvent:*

$$S_\eta(z)^{-1} = P(z - L_\eta)^{-1}P|_{\mathcal{R}}.$$

Its covariant derivative is the sum of the form derivative, the derivative of the metric projection, and the derivative of the complementary inverse. All three terms are continuous uniformly on compact right-half-plane sets.

Proof. Boundedness of P on V gives the topological decomposition $V = \mathcal{R} \oplus QV$. Coercivity on QV gives the unique complementary solution by Lax–Milgram. Blocking the weak resolvent equation in these two subspaces yields the Schur formula. If $S_\eta(z)r = 0$, then the corresponding full weak solution has zero forcing and hence is zero; thus $r = 0$. Finite dimensionality gives invertibility. Differentiate the Lax–Milgram equation for $W_\eta(z)$ and the definition of the metric projection. The type-(B) bounds in Theorem 1.3 control every term. $\square$

1.4. Model-specific pressure rigidity. For the Sinai platform, let $\mathcal{V}_{\text{Sin}}$ be the A4 Hölder-weighted potential space on the exact history shift. For the hard-sphere platform, let $\mathcal{T}_{\text{HS}}$ be the B3 tangent graph space. Pressure functionals are those already constructed in A4 and B2/B4.

Lemma 1.5 (Countable-history Livšic theorem). *If $F \in \mathcal{V}_{\text{Sin}}$ has zero sum on every regular periodic history, then there is $u \in \mathcal{V}_{\text{Sin}}$ with $F = u - u \circ \sigma$. The coboundary range is closed in the weighted norm.*

Proof. Fix a regular reference history x_* . For a history x in its homoclinic class, close a length- n central word by a connector of uniformly bounded cost and define $u(x)$ as the limit of the partial sum of F relative to the closed reference word. Exponential variation makes successive definitions differ by $C\theta^n$, so the limit exists and has the same weighted Hölder norm. Periodic zero sums make it independent of the connector. Mixing makes the homoclinic class dense, and the Hölder bound extends u uniquely. The construction is a bounded linear right inverse of $u \mapsto u - u \circ \sigma$ on the zero-periodic-sum subspace, proving closedness. $\square$

Lemma 1.6 (Hard-sphere annihilator). *A continuous pair of density/contact sources (ϕ, ψ) annihilates every smooth B3 balanced tangent if and only if*

$$(\phi, \psi) = (-\partial_t r - v \cdot \nabla_x r, \Delta r) + \zeta,$$

where r lies in the adjoint graph domain and ζ is a linear combination of the exact mass, momentum, energy, and endpoint constraint functionals. The annihilator is closed.

Proof. The “if” direction is the weak balance identity. Conversely, quotient by the finite-dimensional constraint functionals and apply B3’s closed-range theorem to the linearized balance operator. The annihilator of its kernel is the range of the adjoint because the range is closed. Thus the source is $\mathcal{B}^*r = (-\partial_t r - v \cdot \nabla_x r - \mathcal{L}r, \Delta r)$; the collision part of the adjoint equation absorbs $\mathcal{L}r$ into the linearization of the density source, giving the displayed gauge pair. The closed range and finite-dimensional constraint space prove closedness. $\square$

Theorem 1.7 (Pressure rigidity). *Suppose two regular potentials have identical perturbed pressure on a neighborhood of every strong source.*

- (i) *On the Sinai platform their difference is a constant plus a closed dynamical coboundary.*
- (ii) *On the hard-sphere platform their difference is a balance gauge plus collision invariants and endpoint constraints as in Lemma 1.6.*

Proof. Differentiate the pressure identity in every source direction. Analyticity from A4 and B2 identifies all equilibrium expectations and hence every periodic-orbit derivative in the Sinai core and every balanced tangent pairing in the hard-sphere chart. Subtract the common mean to remove the constant. Lemma 1.5 gives the Sinai coboundary. Lemma 1.6 gives the

hard-sphere gauge. The closedness statements prevent passage to a larger informal closure. $\square$

1.5. Changing filtrations and the innovation representation. Let $(X^\varepsilon, Y^\varepsilon, L^\varepsilon)$ be exact microscopic state, observation, and likelihood processes, and let $\mathcal{G}_t^\varepsilon = \sigma(Y_s^\varepsilon : s \leq t)$. Put

$$M_t^\varepsilon = \mathbb{E}[L_T^\varepsilon \mid \mathcal{G}_t^\varepsilon].$$

Assume the A3/B3 tightness conditions, the C1 regular conditional-kernel convergence, and $\sup_\varepsilon \mathbb{E}|L_T^\varepsilon|^{1+\delta} < \infty$.

Theorem 1.8 (Extended weak convergence of optional projections). *The quadruples*

$$(X^\varepsilon, Y^\varepsilon, L^\varepsilon, M^\varepsilon)$$

are tight in the product Skorokhod topology, and every limit satisfies

$$M_t = \mathbb{E}[L_T \mid \mathcal{G}_t], \quad \mathcal{G}_t = \sigma(Y_s : s \leq t)^{\text{rc}}.$$

Finite-dimensional conditional laws converge jointly at every collection of continuity times, and $M^\varepsilon \Rightarrow M$ in process law.

Proof. Approximate L_T in L^1 uniformly by bounded continuous functions of a finite collection of future observation cylinders. C1's kernel convergence then gives convergence of their conditional expectations jointly at finitely many times. Uniform $L^{1+\delta}$ integrability removes the approximation.

For process tightness, Doob's inequality gives compact containment. For stopping times τ_ε and $h \leq \eta$, write the increment as the conditional expectation of the part of the cylinder approximation lying in $(\tau_\varepsilon, \tau_\varepsilon + h]$ plus its approximation error. The A3/B3 stopping-time modulus and C1 kernel continuity make the first term tend to zero uniformly; the second is uniformly small. This is the Aldous condition, not merely a maximal bound. Right-continuous augmentation and a monotone-class argument identify the limiting conditional expectation. $\square$

In the Gaussian limit supplied by A4 or B3, write the observation as

$$dY_t = b_t dt + \sigma_t dB_t,$$

with uniformly nondegenerate observed covariance. Define the innovation

$$I_t = \int_0^t (\sigma_s \sigma_s^*)^{-1/2} (dY_s - \hat{b}_s ds).$$

Lemma 1.9 (Innovation martingale representation). *The process I is a Brownian motion in the observation filtration, and every square-integrable observation martingale has a unique representation $N_t = N_0 + \int_0^t H_s dI_s$.*

Proof. The conditional drift makes I a continuous local martingale with bracket tI . Lévy's characterization gives Brownian motion. Nondegeneracy and the C1 filter imply that the observation filtration equals the completed innovation filtration: the observation is reconstructed from the innovation by

the pathwise unique filtering equation, while the innovation is a functional of the observation. The Brownian martingale representation theorem then gives the claim. $\square$

Theorem 1.10 (Projected stochastic exponentials, Girsanov, and BSDE limits). *Let M be the optional-projection limit in Theorem 1.8. Then*

$$\frac{M_t}{M_0} = \mathcal{E} \left(\int_0^t H_s dI_s \right)$$

for the unique innovation integrand H . The stochastic logarithms and brackets of M^ε converge to those of M . Consequently bounded Girsanov transforms and Lipschitz risk-sensitive BSDEs driven by the observed filtration commute with the microscopic limit.

Proof. A positive uniformly integrable martingale has stochastic logarithm $dN_t = dM_t/M_t$ after localization. Lemma 1.9 represents N as an innovation integral, giving the exponential formula. The conditional-kernel proof above, applied also to products of likelihood increments, gives convergence of predictable brackets. Uniform $L^{1+\delta}$ bounds prevent loss of mass under Girsanov. For a Lipschitz BSDE, subtract the microscopic and limiting equations, use the bracket convergence and Burkholder–Davis–Gundy inequality, and apply Gronwall. Approximation by bounded terminal cylinders and uniform integrability remove the localization. $\square$

1.6. A nonvacuous contraction calculus. A typed contraction consists only of concrete data:

- (i) a continuous state map $T : E \rightarrow F$ with $W_F \circ T \leq CW_E$;
- (ii) exact microscopic maps T_ε with

$$\limsup a_\varepsilon^{-1} \log P(d(T_\varepsilon X_\varepsilon, TX_\varepsilon) > \eta) = -\infty;$$

- (iii) pullback of strong sources $T^* : G \mapsto G \circ T$;
- (iv) a bounded map between the verified form domains commuting with the state maps.

No pressure, rate, covariance, or memory conclusion is part of this definition.

Theorem 1.11 (Verified contraction functor). *For every typed contraction above:*

- (i) *good rates contract by $I_F(y) = \inf_{Tx=y} I_E(x)$;*
- (ii) *pressures obey $Q_F(G) = Q_E(T^*G)$;*
- (iii) *where the pressures are twice differentiable, covariance and cotangent forms push forward by T and pull back by T^* ;*
- (iv) *if the form-domain map is block unitary on the resolved/complementary splitting, the Schur memory transform commutes with contraction.*

The A3 collision-to-physical clock map, the B2 density/contact projections, and the C1 observation/belief map satisfy the hypotheses and are closed under composition.

Proof. The exponentially good approximation and exponential tightness give the extended contraction principle, proving (i). At finite volume the contracted observable is exactly the pullback, so the exponential moments are equal; pass to the locally uniform pressure limit to obtain (ii). Locally uniform analyticity permits differentiation of that identity twice, proving (iii). For (iv), block-unitary conjugacy carries the complementary inverse and the Schur form in Theorem 1.4 to their contracted counterparts.

For A3, the retained terminal prefix makes the two clock maps continuous on rate sublevels. For B2, marginalization of density/contact currents is continuous in the weighted topology and the microscopic map is exact. For C1, the observation and Bayes maps are continuous on the reachable chart and the null convention is explicit. The weight, source, and form bounds were proved in the respective papers. Composition multiplies the weight constants and adds two superexponentially small errors, so the class is nonempty and closed under composition. $\square$

Theorem 1.12 (Round-twenty-one C2 closure). *The cotangent/contraction platform has an exact weighted strict dual, actual invariant-measure coboundary separation, verified type-(B) form transport and Schur compression, model-specific Sinai and hard-sphere rigidity, a changing-filtration optional-projection theorem with innovation representation, and a nonvacuous contraction calculus whose conclusions are proved rather than assumed.*

Proof. Combine Theorems 1.1, 1.3, 1.4, 1.7, 1.8, 1.10, and 1.11. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Twenty-One dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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